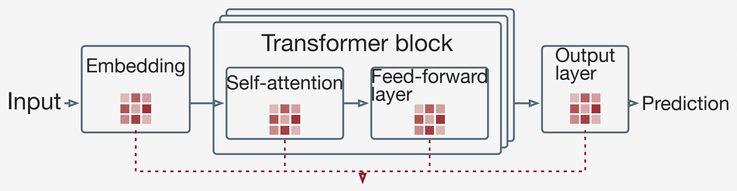
# Adaptive and Neutral Theory of Evolution Strategies for Large Language Models

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## Background LLM fine-tuning with evolution strategies (ES)

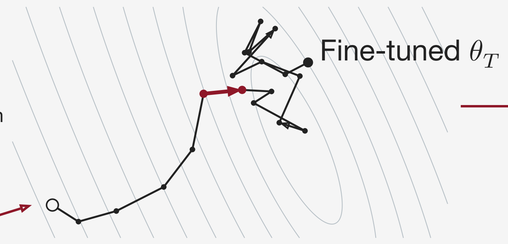
### LLM parameters




All trainable parameters in one vector

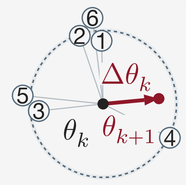
$$\theta = \left(\theta^{(1)}, \theta^{(2)}, \cdots, \theta^{(D)}\right) \in \mathbb{R}^D$$

### LLM parameter trajectory




Reward $R_\sigma$

### Sampling and one update




Perturbed LLM

$$\theta_k + \sigma\varepsilon_n$$

Isotropic perturbations

$$\varepsilon_n \sim \mathcal{N}(0, I_D)$$

$$\Delta\theta_k = \alpha\hat{g}_N = \frac{\alpha}{\sigma N}\sum_{n=1}^{N} w_n \varepsilon_n$$

Rewards only → no backpropagation; less memory

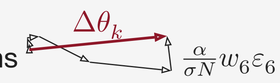
### Response scores and weights

Individual → Reply → Score $R_n$

Z-score: $w_n = \frac{R_n - \text{mean}}{\text{std. dev.}}$



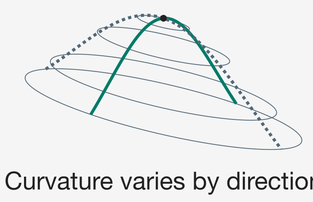

Sum contributions




## Prior work Curvature

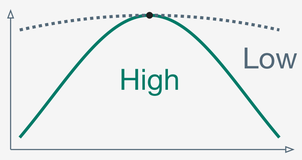
LLM reward landscapes: curvature concentrates in a few directions
Liang et al., 2026

**Reward landscape $R_\sigma$**

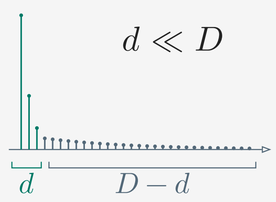


Curvature varies by direction

**Directional sections**




**Curvature $|\lambda_i|$**




## Questions Two puzzles

$N = 30$ individuals can **train all** $D \approx 3 \times 10^9$ **parameters**

Training reward can **decline** as training continues

## Formulation A unified analysis of learning and performance changes

### 1 Stochastic differential equation (SDE)

$$\theta_{k+1} = \theta_k + \alpha\hat{g}_N(\theta_k)$$

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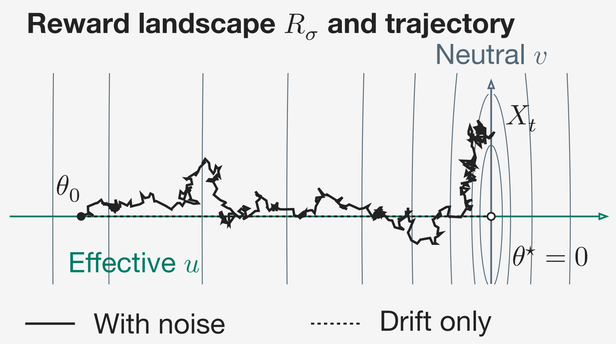
Continuous-time approximation: $X_0 = \theta_0$

$$\mathrm{d}X_t = \underbrace{\alpha b_N(X_t)\,\mathrm{d}t}_{\text{Drift}} + \underbrace{\alpha A_N(X_t)\,\mathrm{d}W_t}_{\text{Noise}}$$

$b_N = \mathbb{E}\hat{g}_N$; $A_N A_N^\top = \mathrm{Cov}(\hat{g}_N)$
$t$: iteration time; $W_t$: standard Brownian motion

### 2 Decompose motion by curvature

**Reward landscape $R_\sigma$ and trajectory**



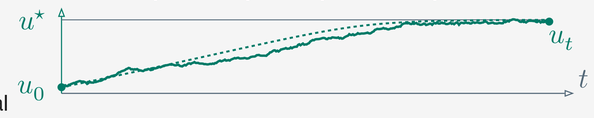

$P$

Orthogonal projection

$I - P$

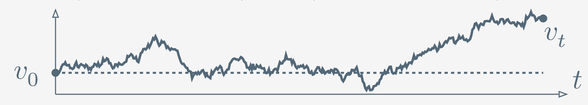
**Effective subspace**

$u_t = PX_t \quad \mathrm{d}u_t = P\,\mathrm{d}X_t$




**Neutral subspace**

$v_t = (I - P)X_t \quad \mathrm{d}v_t = (I - P)\,\mathrm{d}X_t$

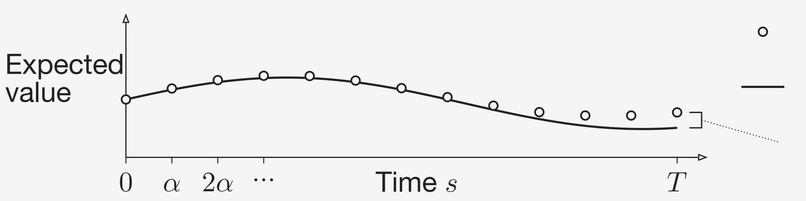



## Main results Finite-time weak approximation and conditions for learning in effective directions and reward decline from fluctuations in neutral directions

### Theorem 1 Finite-time weak approximation of ES

For a test function $\varphi$ and time horizon $T$,

$$\max_{0 \le k\alpha \le T} \left|\mathbb{E}[\varphi(\theta_k)] - \mathbb{E}[\varphi(X_k)]\right| = \mathcal{O}(\alpha) \quad (\alpha \to 0).$$




**Interpretation**

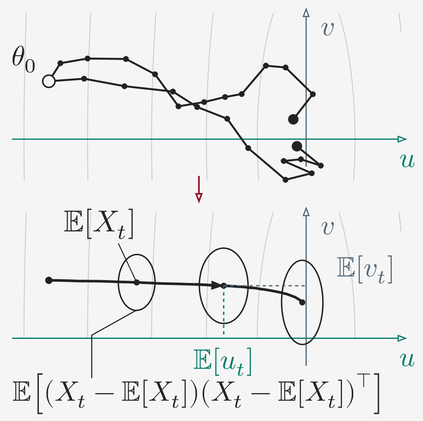
Approximate the mean and covariance of ES with an SDE

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Decompose motion by curvature to assess its effect on reward

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**Study small-population learning and later performance decline**




### Theorem 2 Learning with small populations

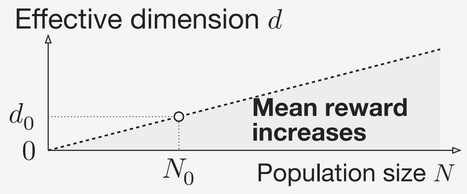
Sufficient condition for mean-reward gain in effective directions

$$\frac{\alpha}{\sigma}\sqrt{2\mu V_u} \quad > \quad \frac{L\alpha^2}{2\sigma^2}\frac{\boldsymbol{d}}{\boldsymbol{N}} \quad + \quad \overline{r}$$

Gain from the mean update — Reward gap $V_u$; local PL $\mu$

Loss due to dispersion — Smoothness $L$

Coefficient approximation error bound




For $d = d_0$ : $N > N_0 = \dfrac{L\alpha d_0}{2\sigma\left[\sqrt{2\mu V_u} - \left(\frac{\sigma}{\alpha}\right)\overline{r}\right]}$

**Interpretation**

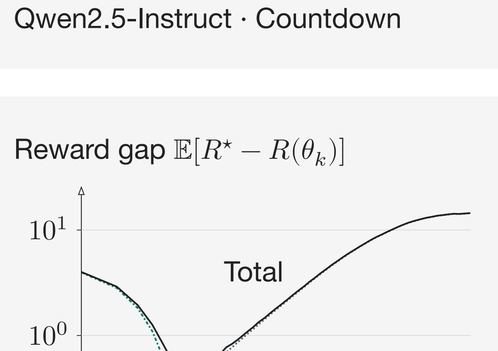
A smaller effective dimension $d$ allows a smaller population $N$

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For LLM reward landscapes, $d$ is much smaller than $D$

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**Small-population ES can train large LLMs**




### Theorem 3 Reward decline as training continues

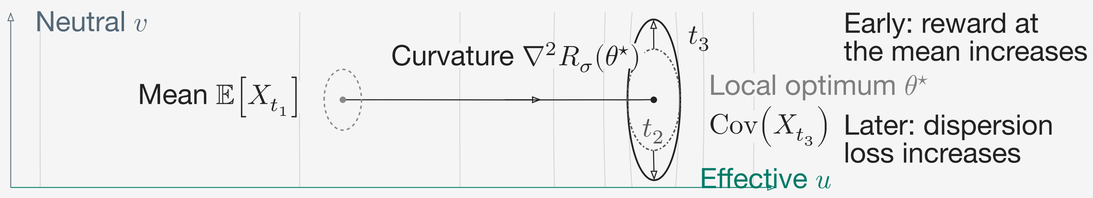
Sufficient for mean training reward $R_\sigma$ to increase $(t_1 < t_2)$

$$\left[R_\sigma(\mathbb{E}[X_t])\right]_{t_1}^{t_2} \quad > -\frac{1}{2}\left[\mathrm{Tr}\left\{\nabla^2 R_\sigma(\theta^\star)\,\mathrm{Cov}(X_t)\right\}\right]_{t_1}^{t_2} + \quad \varepsilon$$

Reward change at mean

Added loss from dispersion

Approximation error bound




Early: reward at the mean increases

Later: dispersion loss increases

**Interpretation**

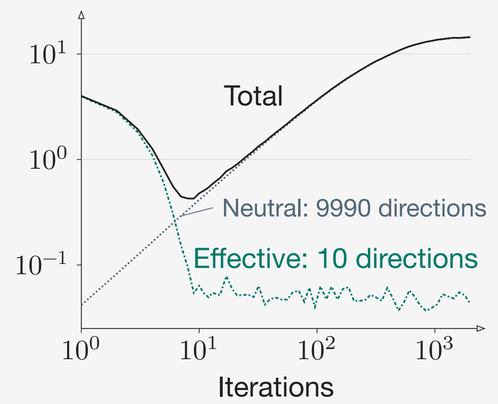
As effective-direction gains slow and neutral dispersion continues

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If the total added loss across neutral directions exceeds the remaining reward gain

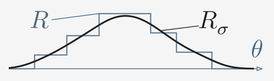
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**Expected reward declines despite further training**




## Outlook Toward compute-efficient fine-tuning and continual learning with less forgetting

**Note**




$R_\sigma(\theta) = \mathbb{E}_\varepsilon[R(\theta + \sigma\varepsilon)]$
$\alpha$: update coefficient
$\sigma$: perturbation scale